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N + - Network Fundamentals & Building Networks.

Introduction to Dynamic Routing.

YAGNIK GHETIYA

TASK 1.

❖ List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

- **Differences Between Static Routing and Dynamic Routing:**

Basis	Static Routing	Dynamic Routing
1. Configuration	Routes are added manually by the network administrator. Like saving a fixed route in Swiggy's delivery system.	Routes are updated automatically using routing protocols. Like Google Maps suggesting routes based on changing road conditions.
2. Network Changes	Does not automatically change routes when a network path fails. Like following a fixed delivery route even when a road is blocked.	Can automatically select another available route when network conditions change. Like Google Maps finding an alternative route during traffic.
3. Resource Usage	Uses fewer network resources because routers do not exchange automatic routing updates.	Uses more network resources to exchange routing information and update routes.

TASK 2.

❖ Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

- **Observing RIP Routes on Three Routers:**

Commands used on each router:

show ip route
show ip protocols

Router	RIP-Learned Routes	Metric
Router 1	Networks learned from Router 2 and 3	Hop count
Router 2	Networks learned from Router 1 and 3	Hop count
Router 3	Networks learned from Router 1 and 2	Hop count

Observation: Routes marked with R in the routing table are learned through RIP. RIP selects routes based on hop count, with a maximum of 15 hops. Exact routes and metrics depend on the network file.

Test Connectivity:

TASK 3.

- **Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route, Use the 'shutdown' command on an interface and monitor with 'show ip route' repeatedly.**

- **Observing RIP Convergence:**

Disable Interface: On one router, the interface was shut down using:

Router(config)# interface serial0/0/0
Router(config-if)# shutdown

Monitor Routing Tables: The show ip route command was repeatedly used on the other routers to check when the unreachable route was removed.

Observation: RIP updated the routing tables after detecting the link failure. The convergence time depends on the topology and RIP timers.

Result:
The failed route was removed after RIP convergence. Exact convergence time should be recorded from the actual Packet Tracer simulation.

TASK 4.

- **Explain in 3-4 sentences how dynamic routing in apps like Uber or Zomato delivery tracking is similar to RIP dynamic routing in networks. Focus on how routes or paths are updated automatically in both cases.**

  - Dynamic routing in networks is similar to delivery tracking in apps like Uber or Zomato. RIP automatically updates routing tables when network paths change, just as delivery apps adjust routes when roads are blocked or traffic increases. Both systems use updated information to find an available path. This helps maintain connectivity in networks and smooth delivery operations in apps.